

GAIRA V7 — Phase 05

The Canonical CSM Inference Engine

Figure report — all 15 figures with captions · Raman only

Status	GATE_FAILED — 15 of 16 gates pass
Replaces	the Phase 04 Theme/BSV inference path
Engine fingerprint	20d8bd99ce71f45a125c6a2b1d719e51
Frozen inputs	LSM + CSM verified, nothing refitted
Representation	49-d frozen CSM activation vector
Similarity metric	cosine (nested grouped CV)
Calibration	dirichlet (selected on Brier)
Split A molecule top-1 / top-5	0.605 / 0.795
Split B class top-1 / top-3	0.845 / 0.971
Split B macro F1	0.807
Calibration ECE / discrimination	0.130 / 0.891
Open-set joint AUROC	0.921 (synthetic negatives)
Evidence axes grounded	7 of 11 (± 1 under the window sweep)
Provenance chains broken	0 of 3133
Robustness retention, CSM vs raw	0.935 vs 0.895 (class, molecule-grouped)
Completed	2026-08-06 20:13:57 UTC

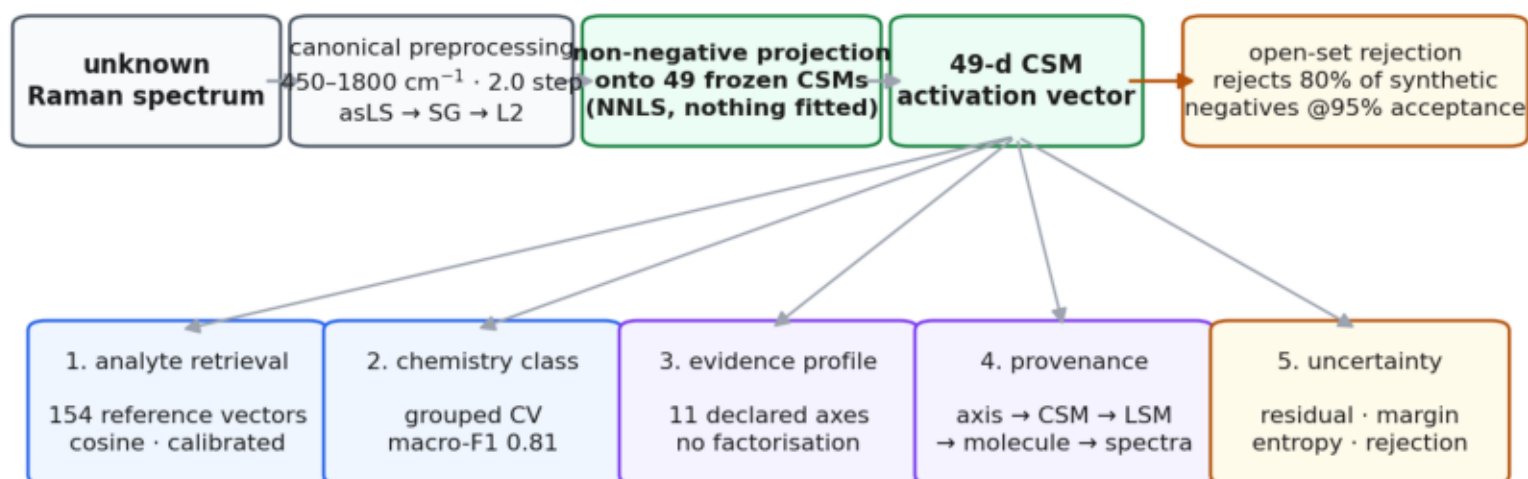
Committed sources of record: reports/PHASE_05_CANONICAL_INFERENCE_ENGINE.md and reports/PHASE_05_SCIENTIFIC_AUDIT.md
G6 (calibration ECE $\leq$ 0.10) FAILS at 0.130. It was not relaxed: on this corpus that threshold is reachable only by a calibrator that reports the same confidence for every spectrum. Figures are embedded from the committed PNGs.

Figures

01 canonical pipeline
02 csm projection
03 class confusion matrix
04 calibration benchmark
05 reliability diagram
06 topk retrieval
07 confidence
08 openset rejection

09 evidence radars
10 provenance waterfall
11 noise robustness
12 representative reports
13 active csms
14 molecule csm evidence
15 architecture summary

Figure 1 • The canonical GAIRA inference pipeline

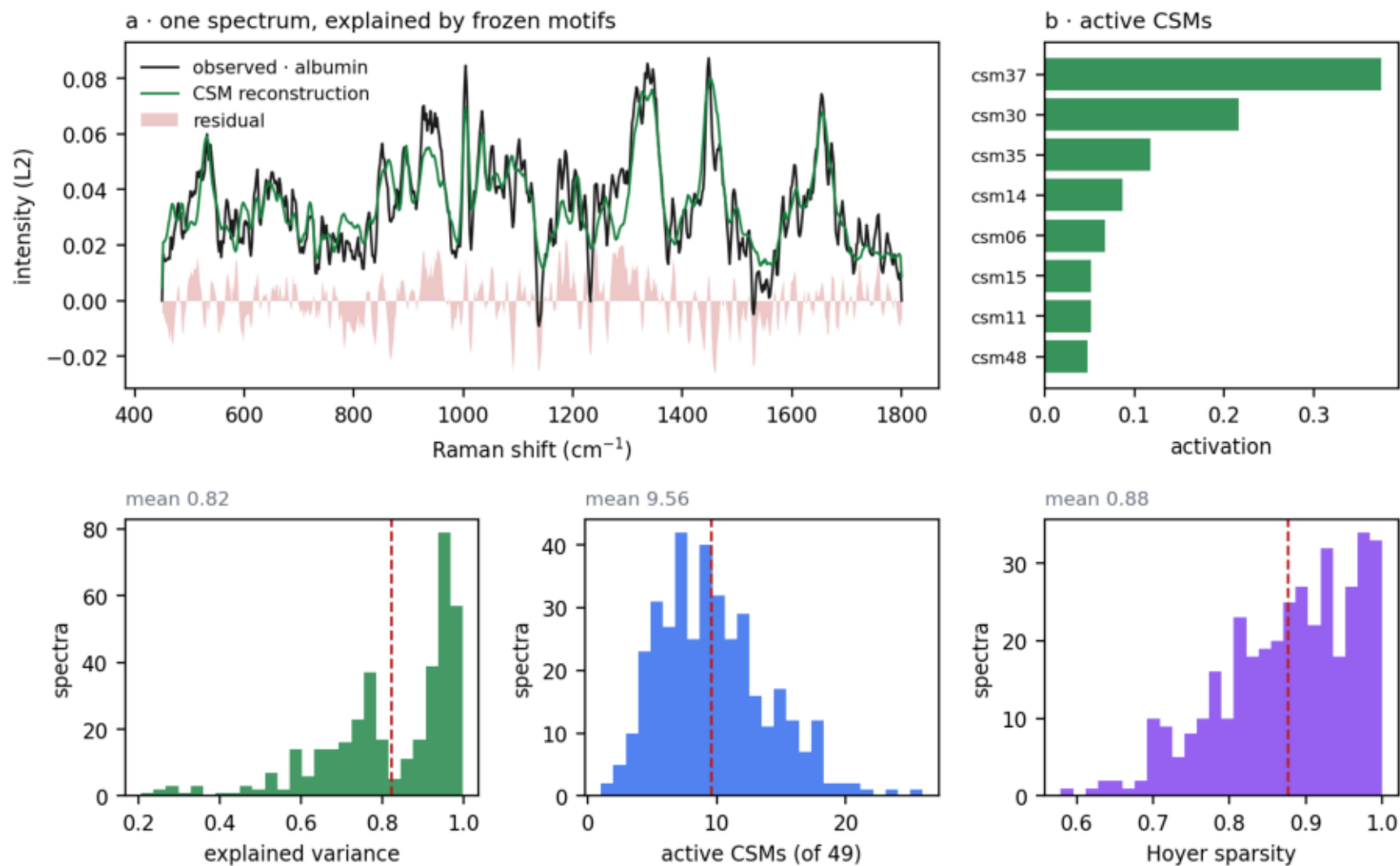


Geometry is visualisation only — no inference path passes through it. Everything above the branches is frozen; the engine fits nothing at inference.

The canonical pipeline. A spectrum becomes a 49-d frozen CSM activation vector and five inference branches read that one vector. Nothing is fitted at inference; geometry appears nowhere in the path.

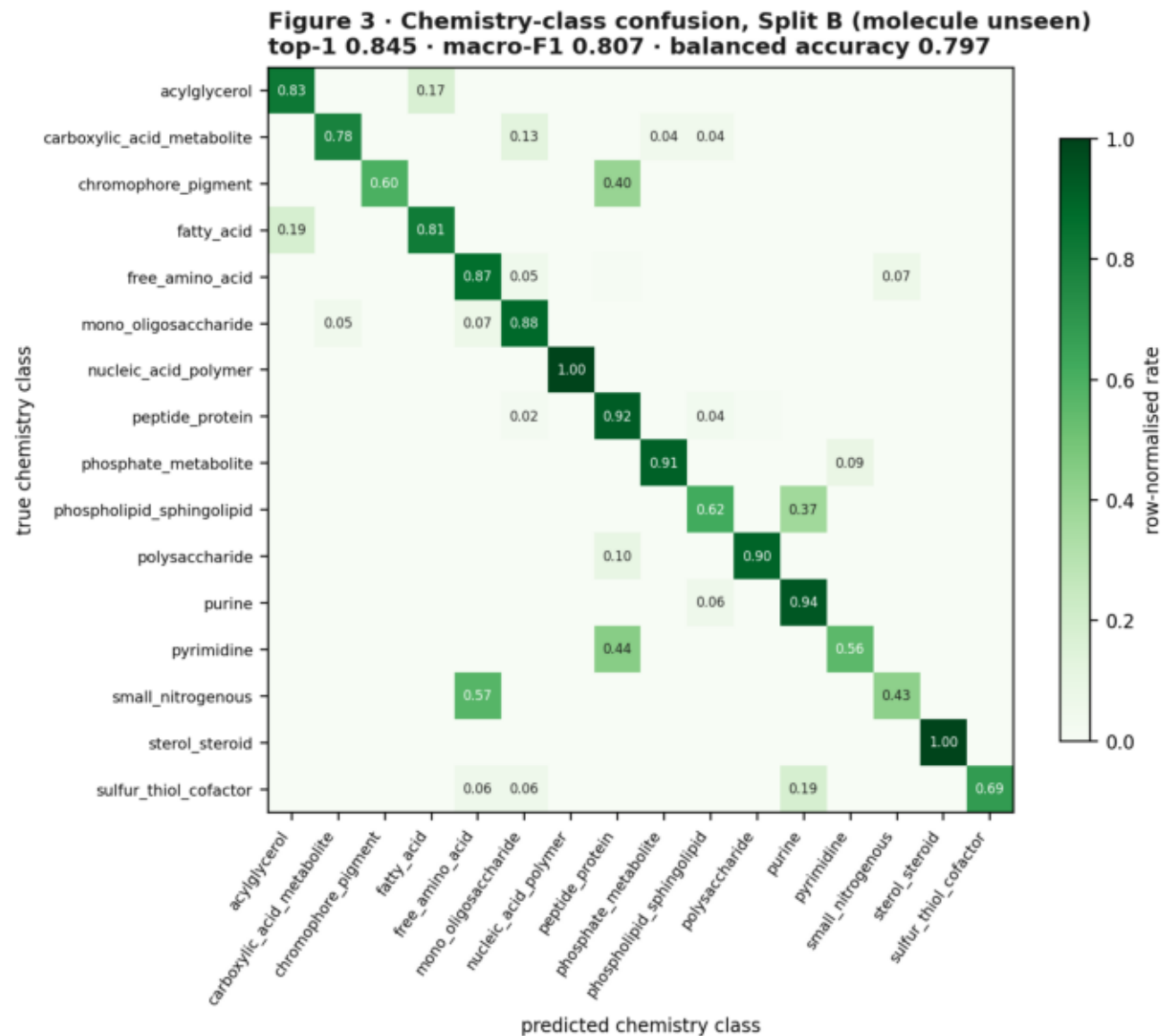
Figure 02

Figure 2 · Direct projection onto the frozen CSM dictionary



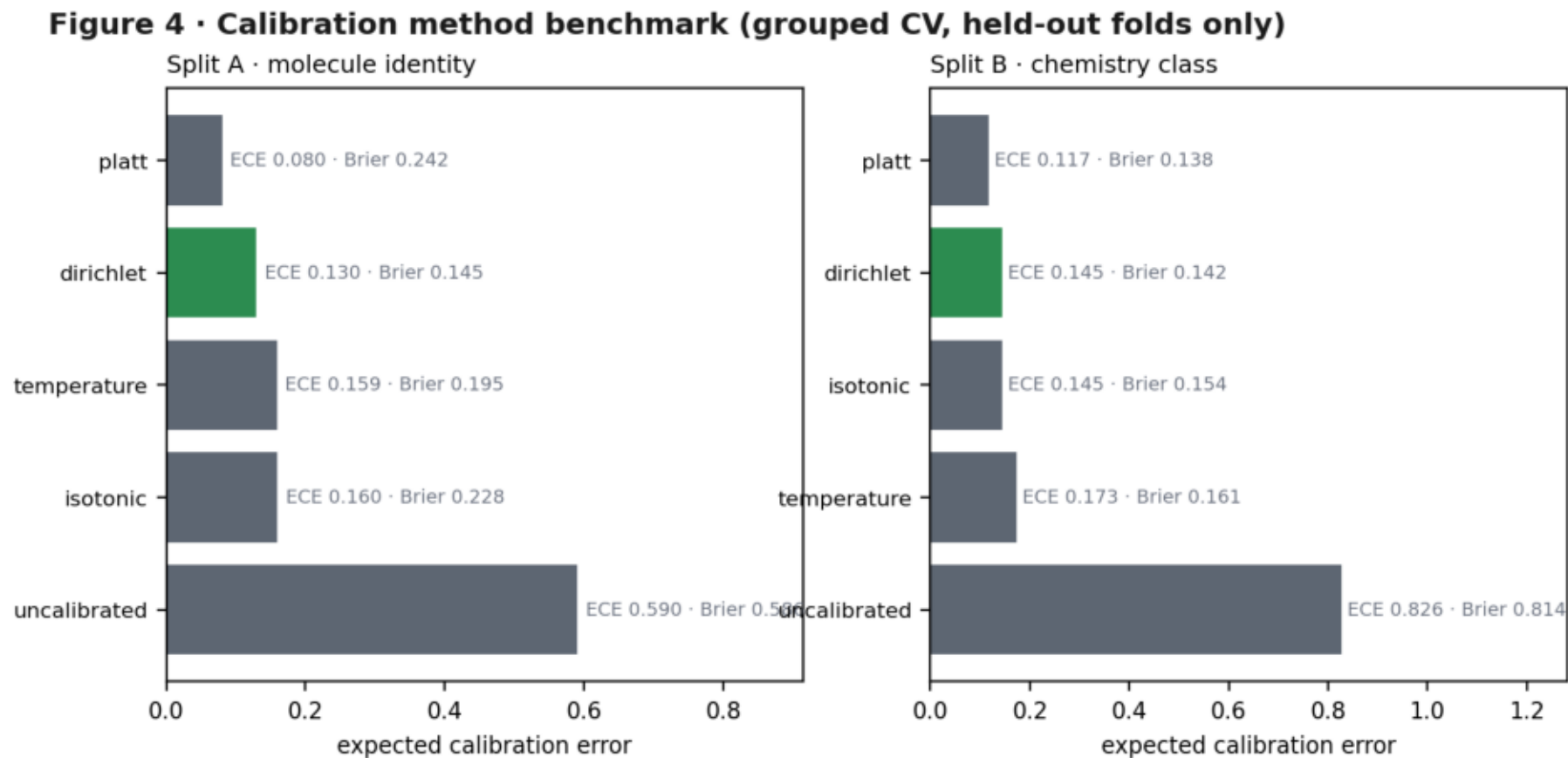
Direct NNLS projection onto the frozen dictionary. Mean explained variance 0.821 with 9.6 of 49 motifs active — the shape a mixture model should have. The minimum of 0.206 is real: those spectra are what the rejection channels exist for.

Figure 03



Chemistry-class confusion on molecules the atlas has never seen (Split B). The failures sit where a spectroscopist would put them — small classes dominated by chains and rings they share with larger ones.

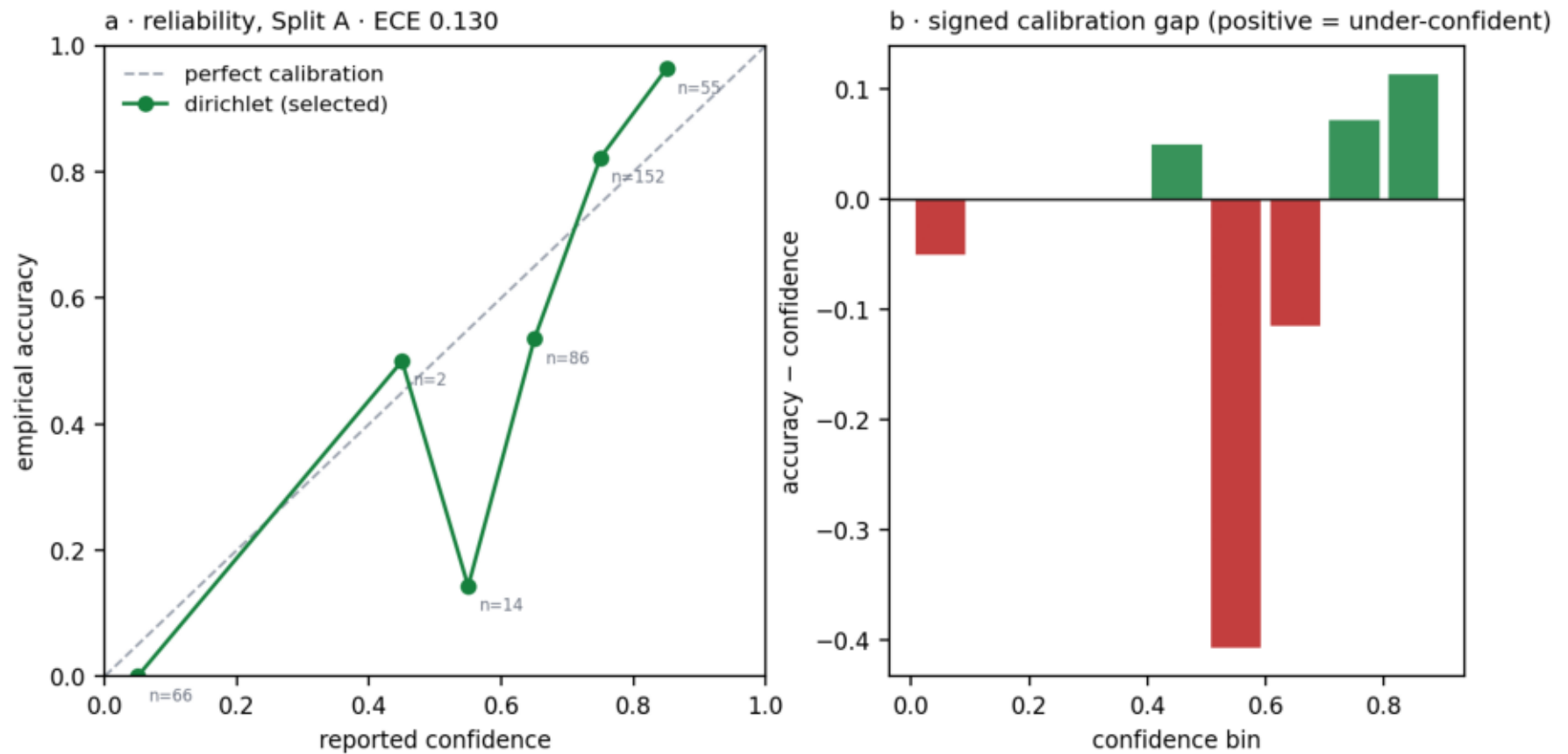
Figure 04



Calibration benchmark. Platt scaling wins ECE (0.080) and loses Brier (0.242) because it reports the base rate for every spectrum. Selection is on Brier, which is proper and cannot be won by flattening.

Figure 05

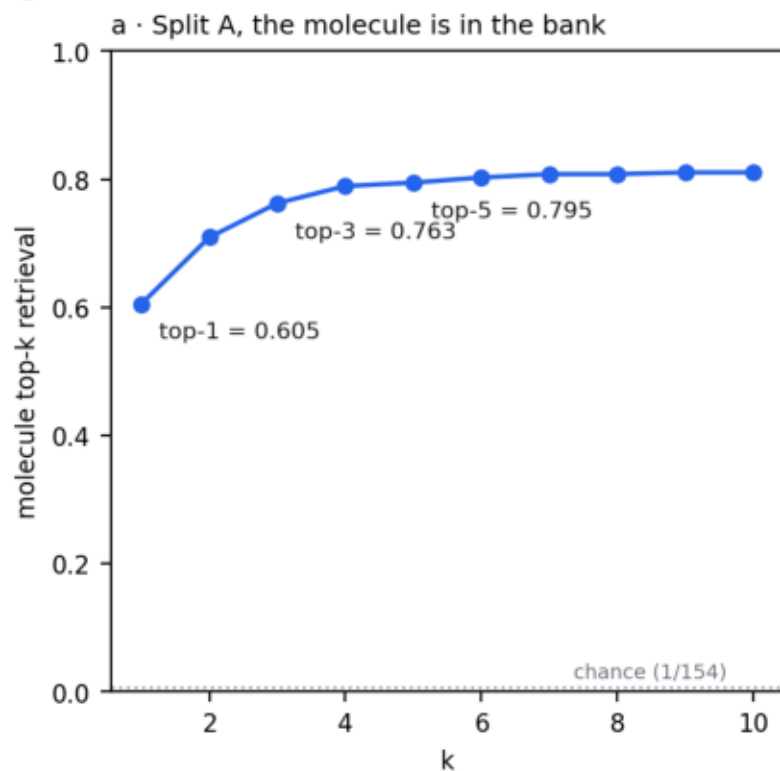
Figure 5 • Reliability diagram



Reliability of the selected Dirichlet calibrator. ECE 0.130 — above the pre-declared 0.10 gate, which is reported as a failure rather than relaxed.

Figure 06

Figure 6 · Retrieval under the two evaluation splits



Split B · the molecule is absent from the bank

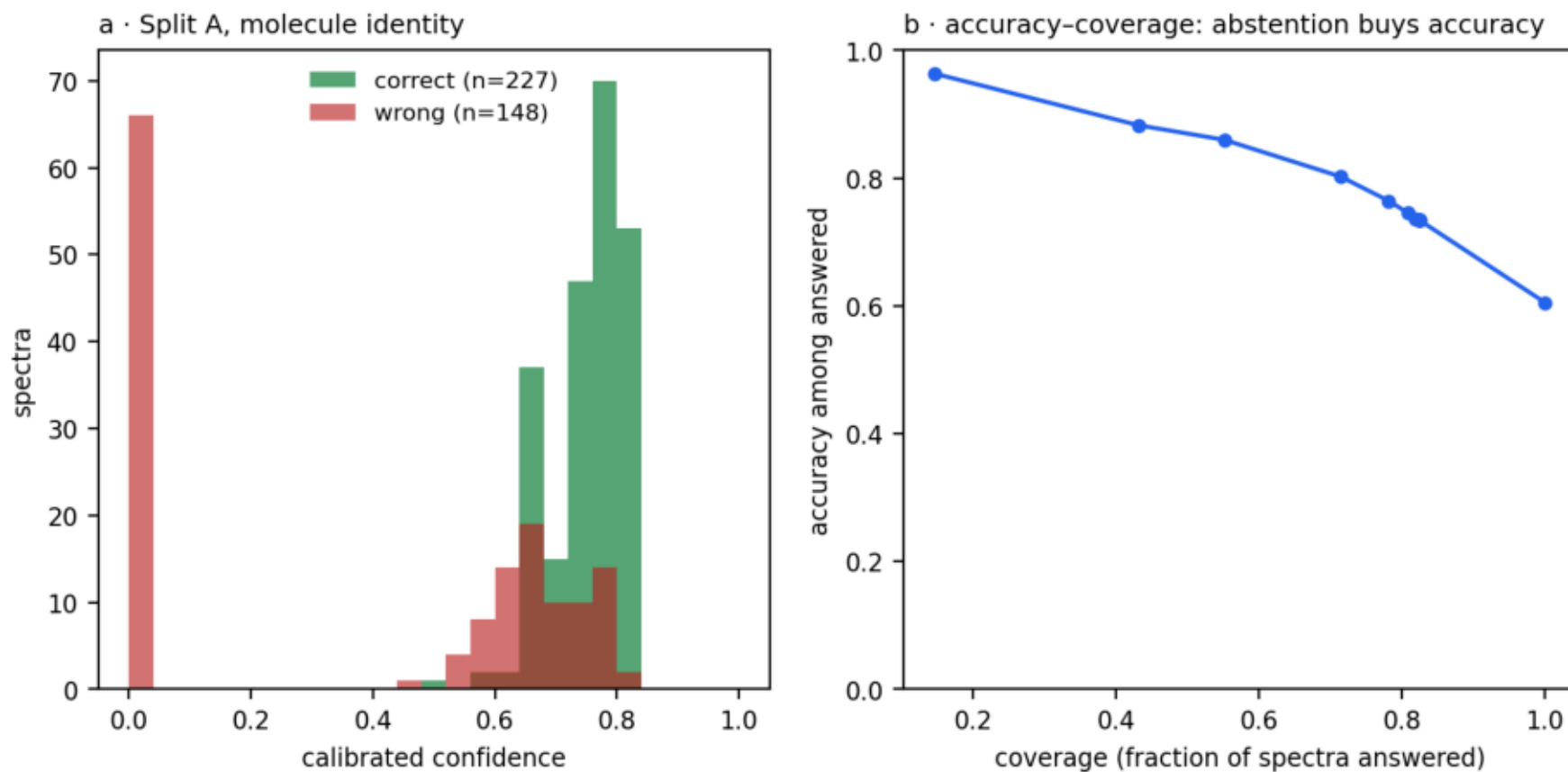
Molecule top-k is **undefined**, not zero: the correct answer is not among the candidates. Reporting 0.000 would be a category error.



Retrieval under both splits. Molecule top-k is undefined under Split B, not zero: the correct answer is not among the candidates.

Figure 07

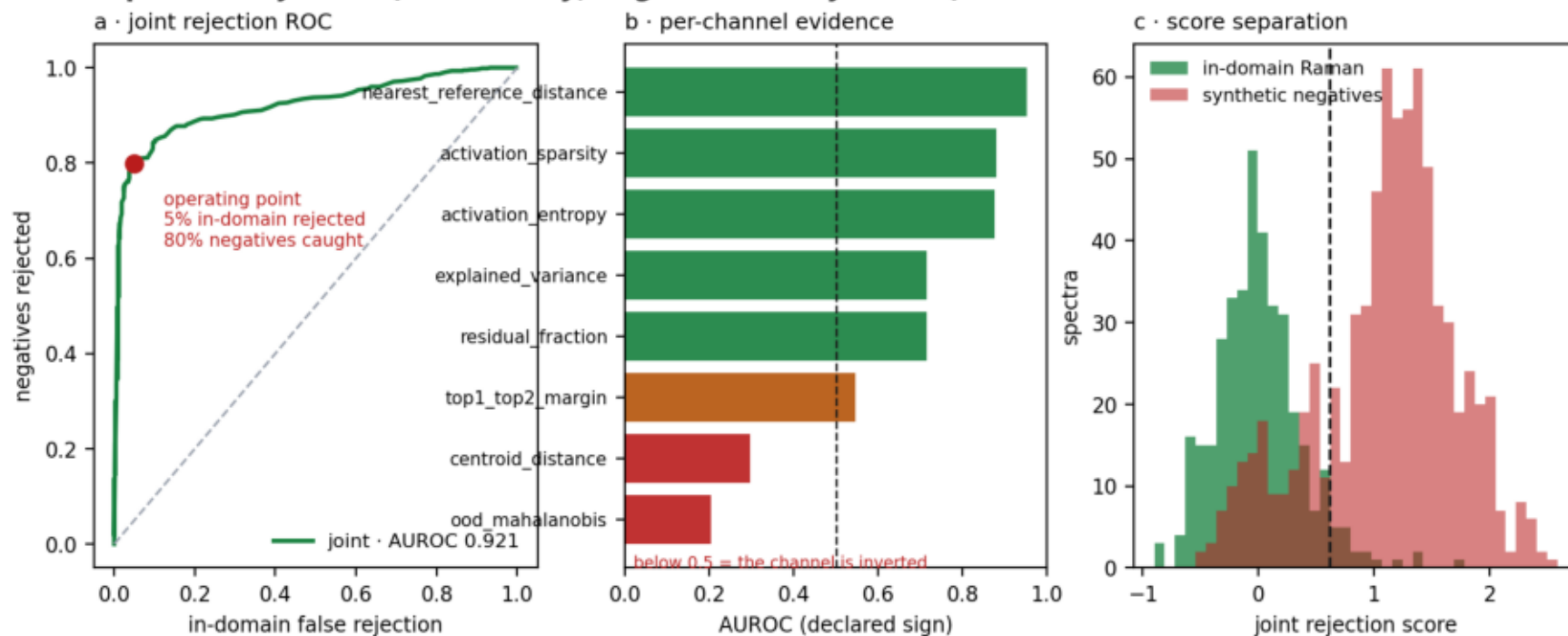
Figure 7 • Confidence behaviour



Confidence separates correct from wrong answers (AUROC 0.891) and thresholding trades coverage for accuracy monotonically — what a usable confidence must do.

Figure 08

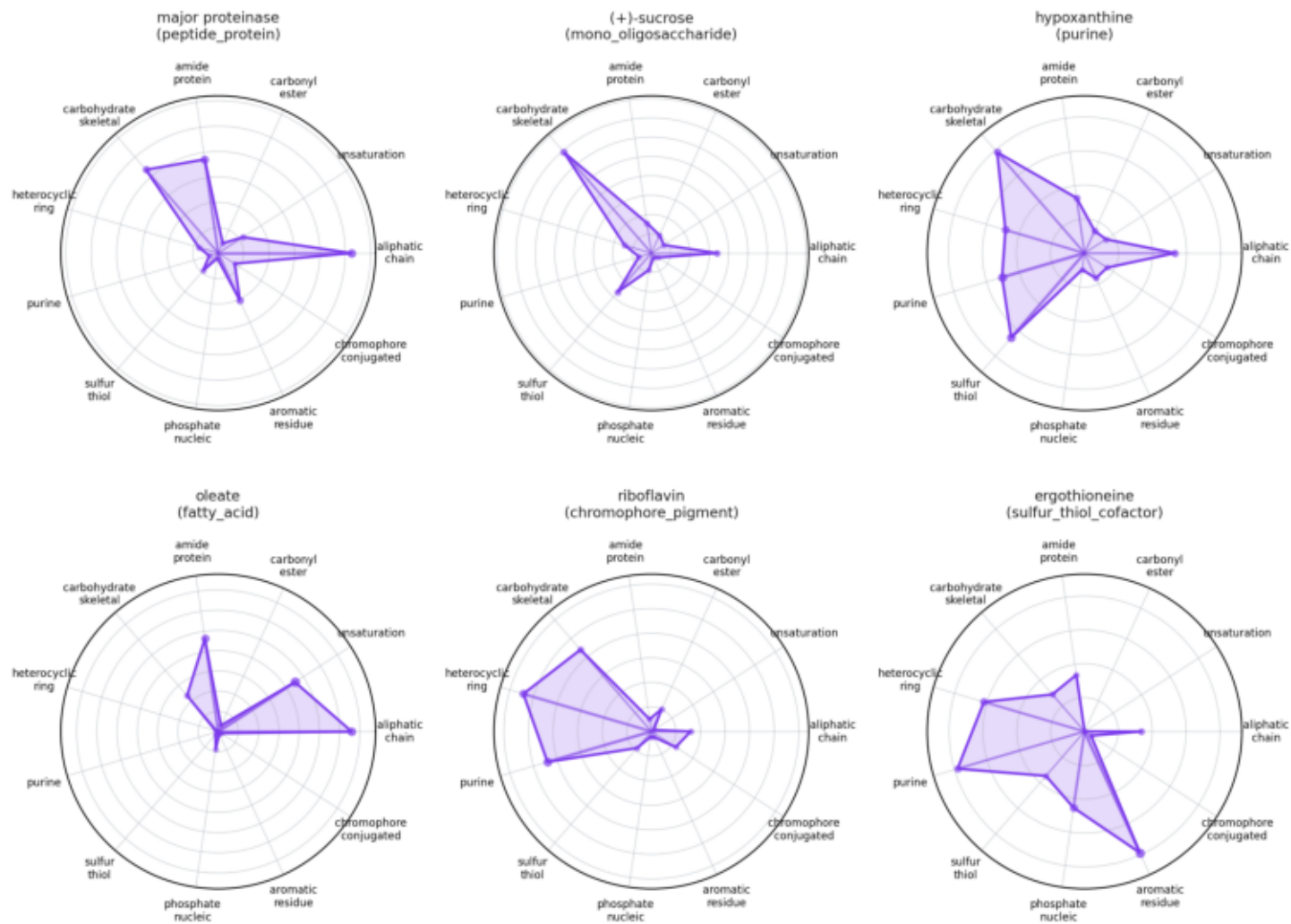
Figure 8 · Open-set rejection (Raman only; negatives are synthetic)



Open-set rejection on synthetic negatives. Two channels score below chance because in a sparse non-negative code the population centre is where degraded spectra land, not where real ones do. Signs are reported unflipped.

Figure 09

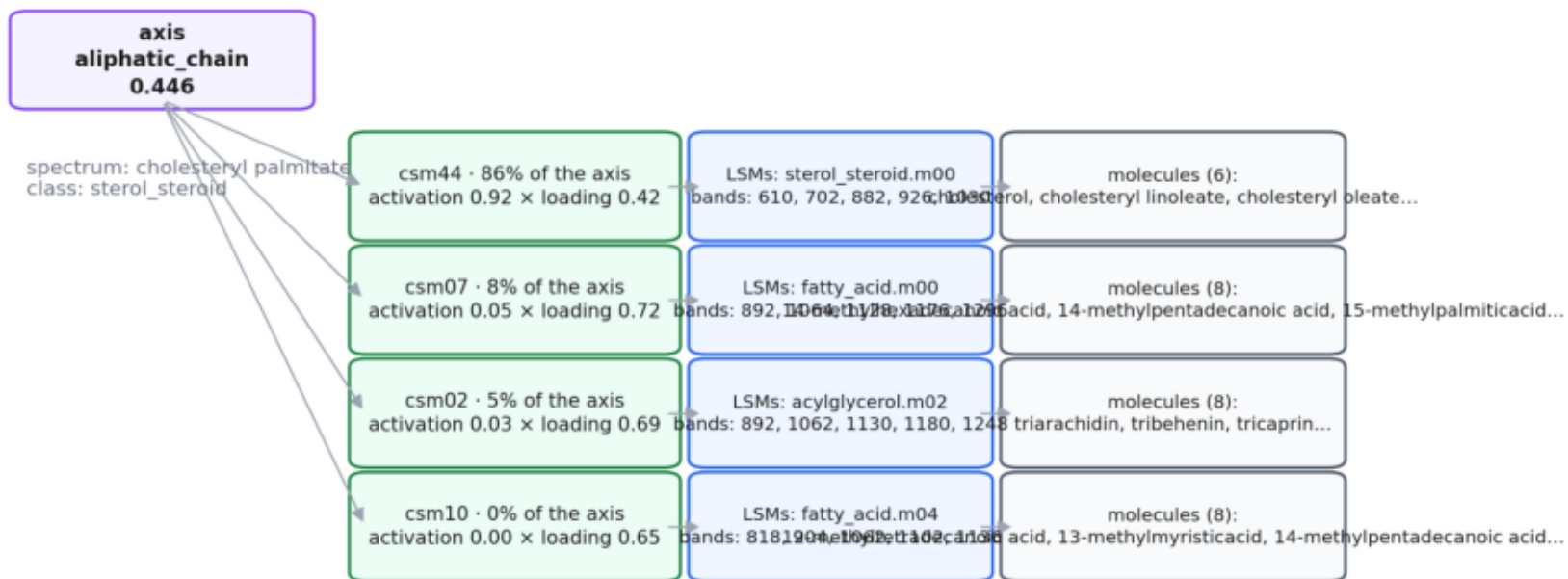
Figure 9 • Biochemical Evidence Profiles
spoke thickness and marker size encode confidence — weak evidence looks weak



Biochemical Evidence Profiles. Eleven declared axes, no factorisation. Spoke thickness and marker size encode confidence, so an axis resting on one motif in a poorly reconstructed spectrum looks thin.

Figure 10

Figure 10 · Provenance waterfall: axis → CSM → LSM → molecule → spectra

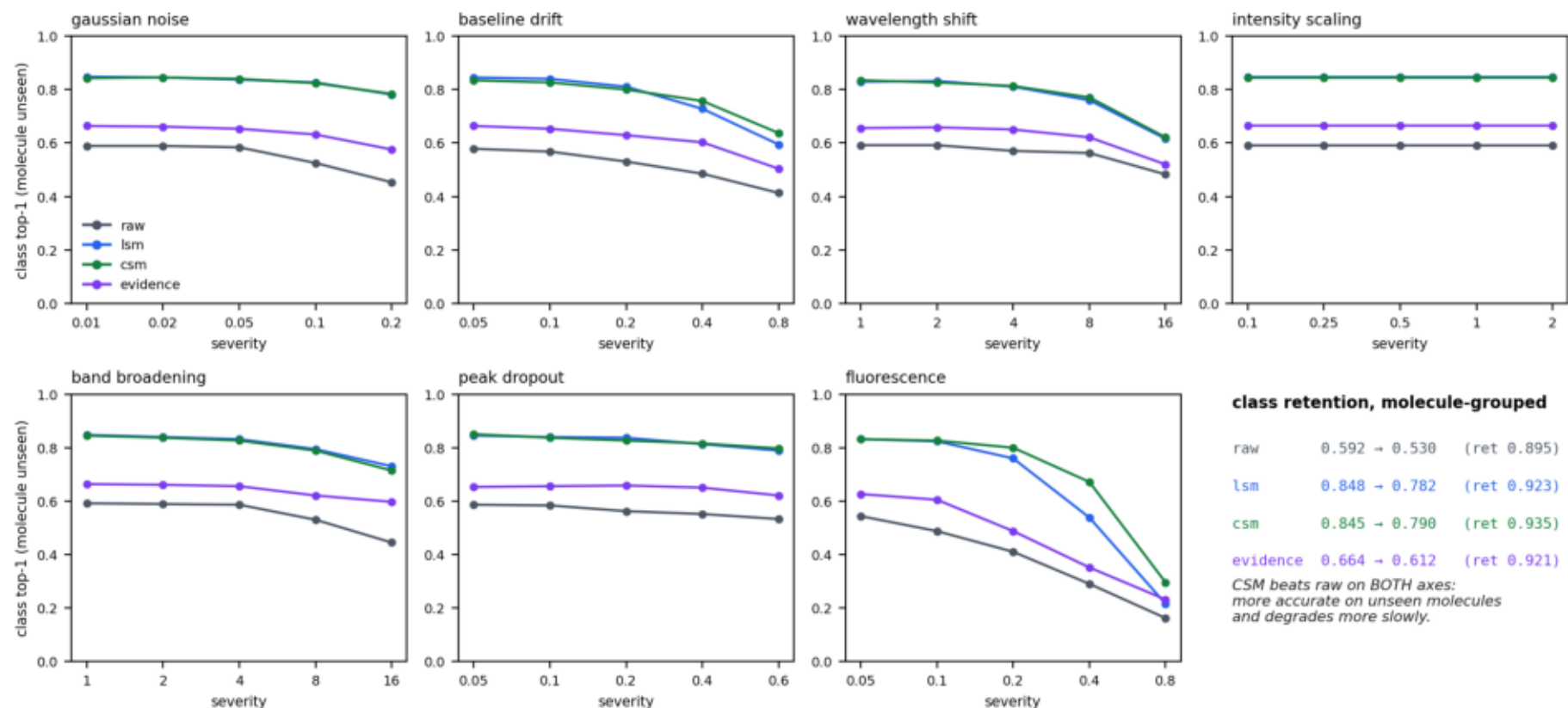


Top 4 of 5 contributing CSMs shown; the listed links account for 100% of the axis value. Every molecule name resolves to measured reference spectra in the frozen corpus.

Provenance waterfall. Contributions are the actual additive terms, so the listed CSMs sum exactly to the axis value. 3,133 chains verified, none broken.

Figure 11

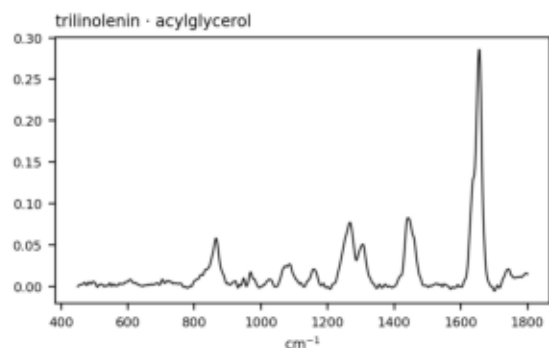
Figure 11 • Noise robustness across representations (chemistry class, molecule-grouped)



Noise robustness on molecule-grouped banks. CSM projection beats the raw spectrum on both halves of the hypothesis: more discriminative on unseen molecules (0.845 vs 0.592) and slower to degrade (retention 0.935 vs 0.895).

Figure 12

Figure 12 • Representative inference reports

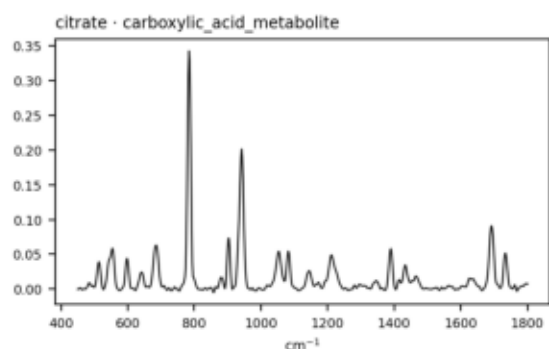


ANALYTE RETRIEVAL
1. arachidonic acid (0.998)
2. trilinolenin (0.997)
3. α -linolenic acid (0.991)

confidence 0.693
margin 0.000
entropy 0.993

CHEMISTRY CLASS
fatty_acid (0.998)

DIAGNOSTICS
EV 0.965 · 10 CSMs
unassigned 0.007

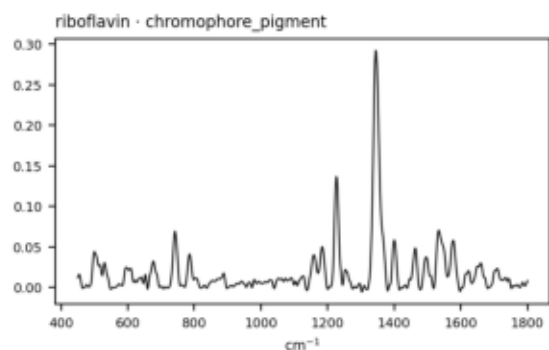


ANALYTE RETRIEVAL
1. citrate (0.998)
2. succinic acid (0.919)
3. pyruvate (0.835)

confidence 0.758
margin 0.078
entropy 0.996

CHEMISTRY CLASS
carboxylic_acid_metabolite (0.998)

DIAGNOSTICS
EV 0.596 · 7 CSMs
unassigned 0.063

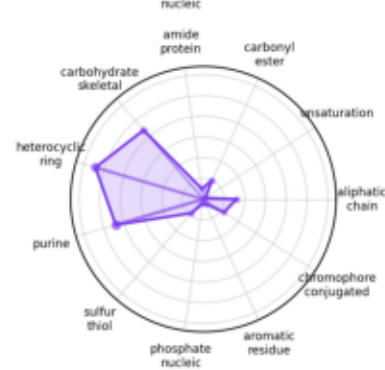
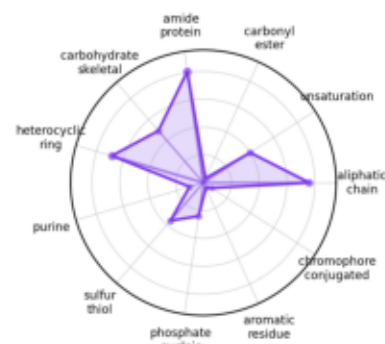
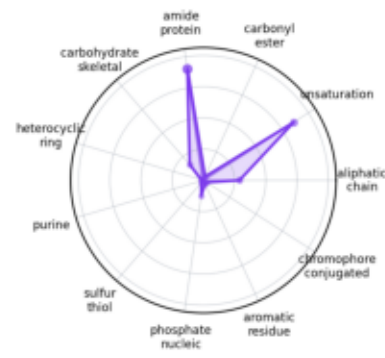


ANALYTE RETRIEVAL
1. riboflavin (0.993)
2. melanin (0.555)
3. glycogen (0.287)

confidence 0.811
margin 0.439
entropy 0.998

CHEMISTRY CLASS
chromophore_pigment (0.993)

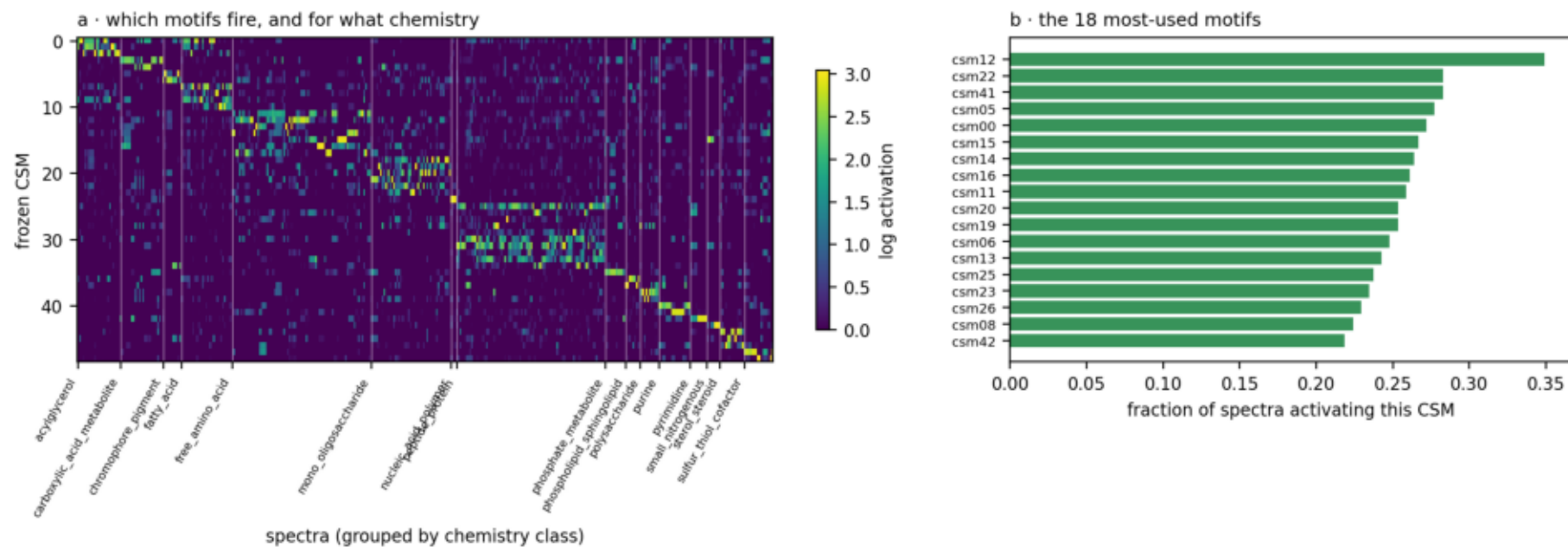
DIAGNOSTICS
EV 0.644 · 10 CSMs
unassigned 0.205



Three complete inference reports: spectrum, retrieval with calibrated confidence, chemistry class, diagnostics and evidence radar. The top case has margin 0.000 — an honest near-tie between two structurally similar lipids.

Figure 13

Figure 13 · Active CSMs across the corpus

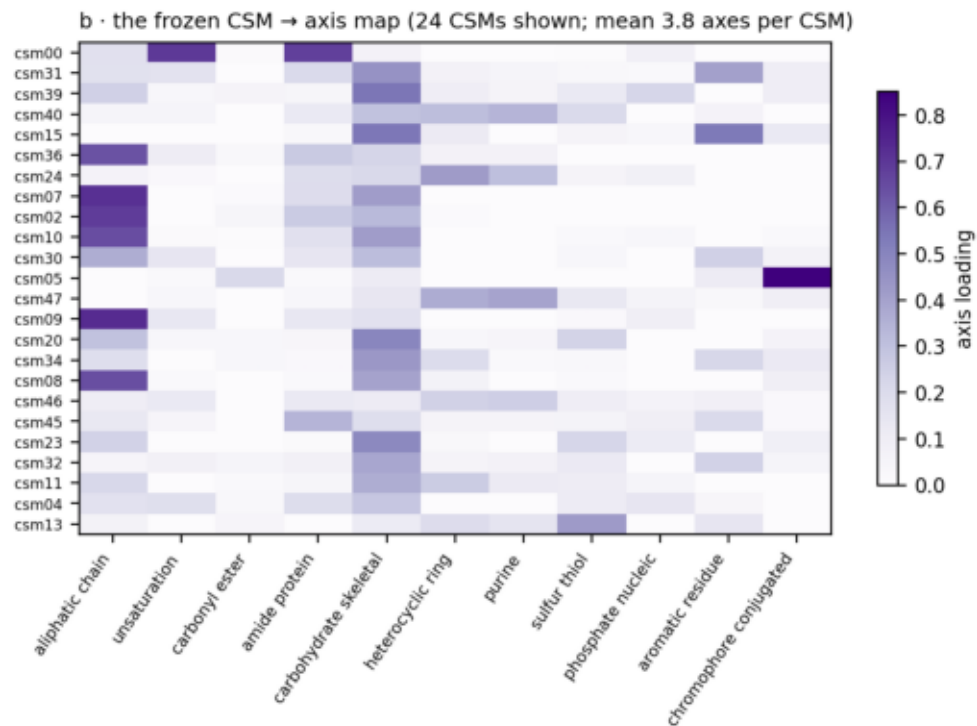
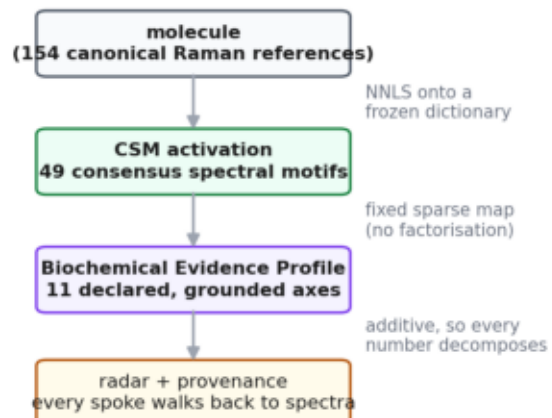


Which frozen motifs fire, and for what chemistry. The block structure is the reason class inference generalises to molecules the atlas has not seen.

Figure 14

Figure 14 · Molecule → CSM → Biochemical Evidence Profile

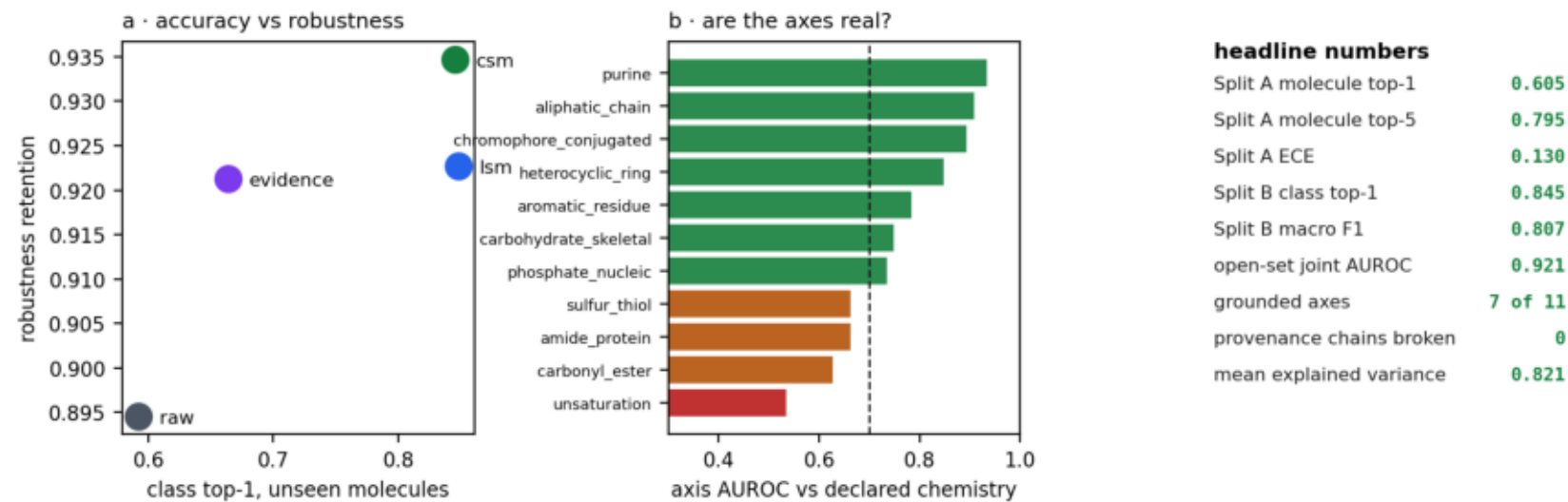
a · the abstraction chain



The abstraction chain and the frozen CSM to axis map. Mean 3.8 axes per CSM; a CSM loads on an axis only when one of its diagnostic bands falls in an axis window.

Figure 15

Figure 15 · Phase 05 architecture summary



G1 frozen fingerprints verified	G2 nothing upstream refitted	G3 CSM projection explains the corpus (mean EV >= 0.80)	G4 Split A molecule top-1 >= 0.60
G5 Split B class top-1 >= 0.60 on unseen molecules	G6 calibration ECE <= 0.10 after selection	G6b calibration is informative (discrimination >= 0.75, sharpness > 0.05)	G7 open-set joint AUROC >= 0.80 on synthetic negatives
G8 at least 6 evidence axes empirically grounded (AUROC >= 0.70)	G8b axis grounding stable across threshold choices	G9 no broken provenance chains	G10 CSM more robust than raw (class retention)
G11 CSM preserves discrimination on unseen molecules (>= raw)	G12 engine deterministic on repeat	G13 no cross-modality experiment in this phase	G14 geometry not used in inference

15 of 16 gates pass · engine fingerprint 20d8bd99ce71f45a125c6a2b1d719e51 · Raman only, no cross-modality experiment

Phase summary. Accuracy against robustness, axis grounding, headline numbers and all sixteen gates — fifteen pass, and the one that fails is argued in section 6.